Girls Build Bots — Brownie Automotive Manufacturing

Badge-aligned workshop guide for robotics-team-led outreach

Level: Brownies (Grades 2–3) · **Badge:** Automotive Engineering — Automotive Manufacturing **Series:** Automotive Engineering (Design · Engineering · Manufacturing) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on craft and cardboard supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Automotive Engineering badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: designing for a driver

The Automotive series follows a car from idea to road: design it, engineer it to move, then **manufacture** it. This final badge asks: once you have one great car, how do you make *hundreds* — the same, fast, and well-made?

Vocabulary: manufacture, factory, assembly line, station, teamwork, quality, inspector.

Automotive Manufacturing (75–90 minutes)

Badge theme: Discovering how cars get made — assembly lines, teamwork, and quality — by manufacturing something together, step by step.

Materials

- Paper-car templates (a simple fold-and-decorate car, identical for everyone)
- Craft supplies split into stations: fold, add wheels (stickers/stamps), add windows, decorate, inspect
- Two timers (phone stopwatches are fine); a “quality checklist” card for the inspector

Activity 1 — One Car vs. a Hundred Cars (10 min)

It’s one thing to build a single model — but a real car company makes thousands, all alike. How? **The assembly line:** instead of one person building a whole car, each person does one job as the car moves past. Henry Ford made this famous.

Activity 2 — Round 1: Everyone Solo (15 min)

Each Scout makes one complete paper car by herself, doing every step. Time it. Notice: it's slow, and the cars come out a little different from each other.

Activity 3 — Round 2: Build the Assembly Line (35 min)

Now set up stations — fold, wheels, windows, decorate, **inspect** — one Scout (or pair) per station. The cars move down the line; each Scout does only her job, over and over. One Scout is the **quality inspector** with a checklist, sending back any car that's missing a part. Time the line. Compare: faster? more consistent? What happened when a station got backed up?

Activity 4 — Factory Debrief (5 min) + Wrap (5 min)

Talk it through: the line made more cars, more alike, faster — but everyone had to work as a team, and one slow station slowed everybody. That's real manufacturing. Count the finished cars; celebrate the team's output.

> **Robot moment (optional):** Share for 60 seconds how the team divides its build season into roles and stations — programming, building, wiring — so lots of work happens at once, just like a factory line.

Completion Checklist — Automotive Manufacturing

By the end of the session, to earn the badge, each Scout can: - [ ] Explain what an assembly line is and why factories use one - [ ] Work one station as part of a manufacturing team - [ ] Compare building solo vs. building on a line (speed and consistency) - [ ] Describe why quality checking matters in manufacturing

---

Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Paper cars and stations are the whole activity. The "robot moment" box is optional.
- **Let the solo round be slow.** The contrast is the lesson — don't rush Round 1.
- **Engineer a bottleneck on purpose.** If the line runs too smoothly, quietly overload one station so Scouts *feel* why balance matters, then debrief it.
- **Rotate the inspector role** if time allows, so more than one Scout owns quality.
- **Celebrate the team total,** not individual cars — manufacturing is a group achievement.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Automotive Engineering badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.